



Society of Petroleum Engineers

SPE-227397-MS

Novel Flow Assurance Technique Using Constructed Blockage Identification Plot Restoring and Gaining Oil Production, South Kuwait

Haya Saeed Faraj AlMuhanna, Kuwait Oil Company, Ahmadi, Kuwait; Ian Adrian Kartawijaya, bp, Ahmadi, Kuwait

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227397-MS](https://doi.org/10.2118/227397-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Thousands of wells in Kuwait Oil Company are flowing through flowline networks to multiple Gathering Centers (GC) while a flow assurance challenge of blockage has been experienced with excessive back pressure, reducing flow rates and downgrading their full potential. For high productivity wells, such unwanted reductions could be substantial in negatively affecting GC production. The huge number of wells to be monitored and analyzed poses a challenge with the limited number of subsurface engineers and field operators to detect and rectify the found blockages. To optimize the blockage detection and rectification workflow, the Subsurface Team from KOC and bp created a practical analytical blockage identifier tool called the BIP (Blockage Identification Plot) that helps to detect anomaly faster among the thousands of producer wells, efficiently helping to avoid potential production loss.

This analysis technique is constructed based on the equations of pressure drop across choke and horizontal pipeline. By populating it using wells production and field data using a constructed cross plot, the suspected blocked wells will be easily screened as high-pressure outliers of the straight-line trend in the cross plot. The plotting formulation is created in which the wells in the same production manifold with normal pressure drop will fall close to the straight-line trend. This enables effective screening of suspected blocked wells and therefore saves time in identifying them amongst the massive well stock.

The technique has been validated in the field and brought great results in increasing overall production. In one of the GCs, from the first campaign the field rectification works resulted around 3,500 BOPD of restoration from cleaning and rectifying identified wells by the BIP. This is an improvement to the production loss detection with an aim to sustain and even increase the GC production. More restored rates are expected in the future with continuous application of the BIP approach

This is the first initiation of a blockage identification workflow for high number of wells network without the need to visit the wells which highly enhance and sustain production and support production operations saving a lot of time and effort. Furthermore, creating screening practices which eliminate production losses due to wells suffering from flow assurance issues.

Introduction

Kuwait Oil Company (KOC) fields are amongst the world's biggest onshore oil fields that have been producing since 1940s. The South and East fields which are currently producing around 1.5 MMBOPD have been the main contributors to KOC oil production. This production comes from a huge number of oil wells, around 1600 wells that are either flowing naturally or assisted by submersible pumps. All of the wells flow through multiple networks of flowlines into the 15 different Gathering Centers (GCs) which are spread across approximately 1000 square km working area.

In the South and East oil fields, the production strategy is to sustain the base production while continuously growing it through new wells, workovers, and rigless wellworks. As the oil fields are getting mature, they experience flow assurance challenges often associated as partial "blockages". It has been observed that some wells may be flowing with excessive back pressure, which subsequently reduces their flow rates and downgrades their maximum rate potential. Given that most wells are producing oil from excellent permeability reservoirs with moderate to high productivity index, such unwanted reduction could be substantial in negatively affecting the GC production. Upon many fields inspection works, it was found out that excessive backpressure could be caused by either: (1) organic and/or inorganic deposition in the surface side, (2) emulsion in the flowing system, or (3) mechanical problem in the choke valve. When categorizing the occurrence point, these blockages could happen in the well choke, causing the upstream choke pressure at wellhead (WH) to increase. The other possible occurrence point is at downstream of the wellhead choke - usually denoted as happening at the flowline (FL). If the well is in subsonic condition, the increase of FL pressure will also increase WH pressure. Other Kuwaiti Field authors, [Al Mohammad and Al-Saqabi \(2016\)](#) performed pipe flow simulations and estimated flowrate loss between 300-8,000 BOPD per well when the modeled flowlines diameters are reduced due to blockage to match the field measured flowing pressure.

Identifying wells with suspected flow assurance problems can be done in detail using full field well and network modeling as in [Al Jadi and Desai et al. work \(2019\)](#). However, as the South and East KOC fields typically have high number of wells with fast-paced workovers and well intervention activities, also with flowline configuration changes, the field model update process is often a big challenge. By itself, the huge number of wells to be monitored and analyzed poses a challenge for detection. Since most of the wells are not equipped with real time sensors, they require manual data readings and manual one-by-one checking. Similarly, the process of blockage verification and rectification by dismantling the WH choke is a manual process. With the limited number of subsurface engineers and field operators to detect and react to the found blockages, some blockages may be found too late.

This situation worsened in the summer period of the year, when the highest ambient temperature record reached 47 degrees Celsius as per noted in [The Global Historical Weather and Climate Data \(2025\)](#). During this period, the available time window on the fields for manual detection and rectification works is limited further by the shortened safe working hours.

With the challenge to optimize the blockage detection workflow, the subsurface team created a practical analytical blockage identifier. The tool, named Blockage Identification Plot (BIP), is constructed based on two known equations in the oil and gas industry, namely the equation of pressure drop in horizontal pipeline and equation of pressure drop across choke. By populating it using GC wells production and field data into the screening tool, the suspected blocked wells will be seen as high-pressure outlier of the straight-line trend. This enables an effective screening of suspected wells and hence saves time in identifying problematic wells. Subsequently, the rectification effort can be optimized to the wells in need, making the effort more effective and time efficient.

Blockage Identification Tool Methodology

In order to quickly identify flow-assurance problematic wells out of the massive well count, while not all wells were instrumented to give real time dataset readings, we introduced a practical modelling analysis to examine all the flowing wells. The technique uses constructed plotting functions, created for both downstream of the choke (Flowline) and upstream of the choke (Wellhead).

Analysis Method for Well's Flowline (Downstream of Choke)

Figure 1 gives an example map of typical one well flowline connection to the GC and the illustration of well pressures measured points of interest, notably the Wellhead Flowing Pressure (WHP) and the Flowline Pressure (FLP).

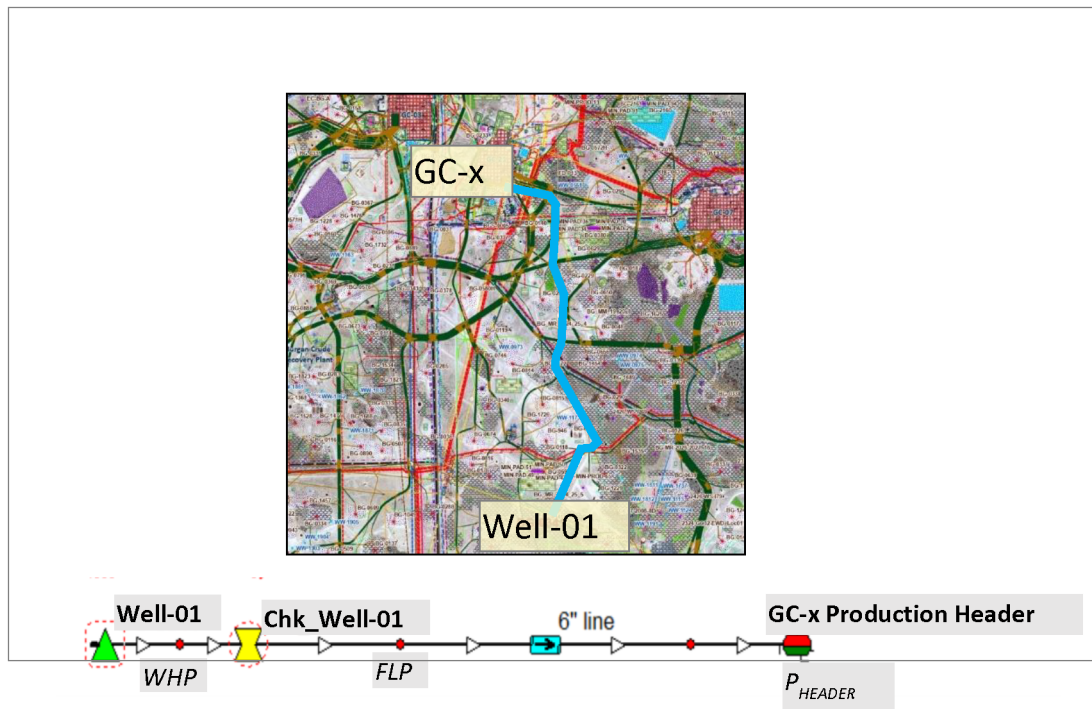


Figure 1—Map Example and Sketch Showing Well Connection Top to bottom: (a) Map Showing Typical Flowline Connection for a Well to GC, (b) Illustration of The Pressure Measurement Points.

The practical modeling for flowline pressure evaluation is based on the general pressure drop equation for horizontal pipe. Note that this equation has been around in many petroleum production engineering textbooks as in Beggs (1991) and Economides (1993).

$$\frac{\Delta P}{\Delta L} = \left(\frac{\Delta P}{\Delta L} \right)_g + \left(\frac{\Delta P}{\Delta L} \right)_f \quad (1)$$

Please refer to the Nomenclature section where each parameter is described.

Assumptions for the South & East Kuwait wells includes:

- Insignificant elevation difference and negligible undulation in the fields. This results in a very small pressure drop from the gravity component.
- All the wells production flowlines in the field are 6". They are installed by the field project team using on-hand stock, with no significant diameter variation.
- In the majority of producing reservoirs, the wells' fluid properties flowing to a particular GC are not too different or have a small degree of variability.
- Header pressures are considered relatively constant.

In horizontal pipeline, the friction pressure loss component of equation (1) is defined as:

$$\Delta P = \frac{f \times L \times v m^2 \times \rho m}{2 \times g \times D} \quad (2)$$

$$FLP - P_{HEADER} = \frac{f \times L \times vm^2 \times \rho m}{2 \times g \times D} \quad (3)$$

Which can be rearranged into:

$$FLP = \frac{f \times L \times vm^2 \times \rho m}{2 \times g \times D} + P_{HEADER} \quad (4)$$

Please refer to the Nomenclature section where each parameter is described.

Examining the equation, for wells flowing with normal pressure drop to the same header in a particular GC, a plot of FLP against the term $(L \times vm^2 \times pm)$ would form a straight line. In this way, we could evaluate the wells without having to history match the friction factor for each well.

The mixture velocity and mixture density can be easily calculated from the well's production test data such as the tested flow rates of oil, water and gas. In the field, well production tests are carried out through means of rigging up portable test separators near the well location periodically (3-monthly to 6-monthly). Post test separator rigged-down, production operator would measure the WHP and FLP several times until next portable test.

For wells with excessive pressure drop, the data points will become high pressure outliers from the straight line in the plot. Figure 2 is an example of the flowline BIP which shows some wells being high pressure outliers. These wells are suspected of having flow assurance or partial blockage issue and would then be communicated to the field team for verification and rectification. Some wells may require flowline flushing using mixture of acid to remove scale, diesel and toluene to remove waxy oil that may be sticking inside the pipe and lastly brine to complete the flush. In addition, to sustain flowline clearance some wells are having chemical injection pumps installation, which continuously pump scale inhibitor to prevent blockage recurrence.

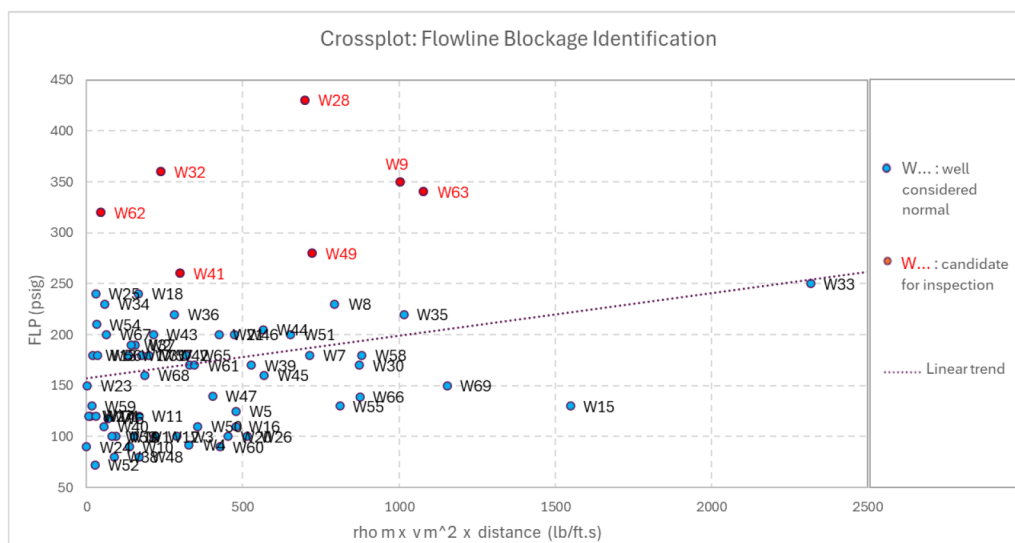


Figure 2—Example of Flowline BIP Showing High Pressure Outlier Wells

Analysis Method for Wellhead (Upstream of Choke)

In South & East Kuwait, the choke types used are positive bean choke and needle and seat adjustable choke. It's quite common for any well to have two wing valves and two choke valves (left and right) open at the same time. For calculation, the equivalent choke size is calculated prior inputting into the BIP.

The practical modeling for wellhead BIP is based on flow equation for choke as presented by various authors such as Gilbert, Ros, or Baxendell. Their works have also been included in many petroleum production engineering textbooks, on which this paper refers to [Beggs \(1991\)](#) and [Economides \(1993\)](#). The empirical base equation for two-phase critical flow through choke is given by:

$$WHP = \frac{A_1 \times Q_L \times GLR^{A_2}}{d^{A_3}} \quad (5)$$

Please refer to the Nomenclature section where each parameter is described.

Examining [equation \(5\)](#), for wells with normal pressure drop for their particular choke opening, a plot of WHP against the term $\left(\frac{Q_L \times GLR^{A_2}}{d^{A_3}}\right)$ would form a straight line. While wells with excessive pressure drop will have their data points become high pressure outliers from the straight-line trend.

This plotting exercise is not trying to simulate the absolute WHP value, but rather for distinguishing wells with normal pressure drop against those with excessive value. It is important to use the consistent set of A_2 and A_3 constants per each author's equation. We have tried plotting using different authors' equations and they all arrived at the same conclusion of screening the wells with excessive pressure drop.

The liquid rate and GLR are taken from the well's production test data. Post test separator rigged-down, production operator would update the WHP and FLP measurements several times until next portable test. Choke size setting is usually kept the same to maintain the same liquid rate post the production test.

[Figure 3](#) is an example of the wellhead BIP which shows some wells being high pressure outliers. These wells are suspected of having flow assurance or partial blockage issue in the choke valve and would then be communicated to the field team for verification and rectification.

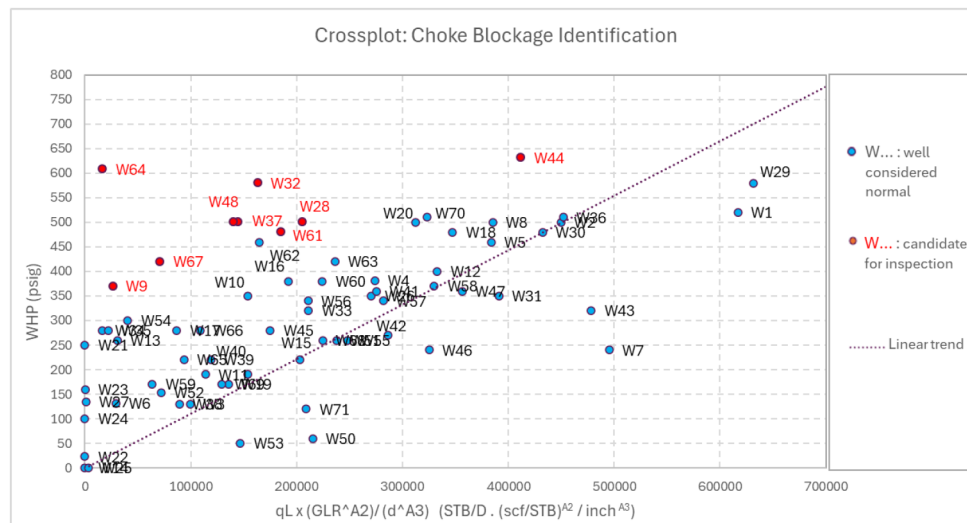


Figure 3—Example of Wellhead BIP Showing Some High-Pressure Outlier Wells

Field Validation

The shown list of wells in [Table 1](#) were identified in the blockage identification plot (BIP) of GC-n wells while reviewing GC by GC in the process of verifying the plots. These wells were sent to the field operation team to confirm the blockage identified by the BIP. Field observation notes included in the "Bean Condition" and "Flowline Spool Condition" columns

Table 1—High Pressure Outlier Wells Sent for Field Check

Well	Last Test QL (BLPD)	Last Test Qo (BOPD)	Bean Condition	Flowline Spool Condition
W9 T	5,749	1,306	10% Blocked, Cleaned	Clear
W121 T	1,557	928	20% Blocked, Cleaned	Found Blockage Required Flow Line Maintenance
W186 T	1,180	407	10% Blocked, Cleaned	Found Blockage Required Flow Line Maintenance
W20 T	900	180	90% Blocked, Can Not Clean It Required Flow Line Maintenance	
			T-Spool Also Blocked, Required Flow Line Maintenance	
			Switched To Elbow Side	
W269 T	1,898	266	30% Blocked, Cleaned	Found Blockage Required Flow Line Maintenance
W152 T	1,358	315	Sample point broken required welding Car to check bean	
W105 T	2,584	853	30% Blocked, Cleaned	
W146 T	800	528	Bean/Spool Clear. Choke indicator incorrect, calibrated: WHP Drop From 270 To 210	
W169	1,703	409	Clear	Clear
			Note: W169 Chemical Injection dosage requires to be increased, emulsion observed	
W192 T	1,843	627	Can't Check Bean, D/S Valve Passing Required Flow Line Maintenance	
W238 T	2,497	1,857	Bean/Spool Clear. Choke indicator incorrect, calibrated: WHP Drop From 580 To 430	
		Sum = 6,734		

The validation process was a success, where most of the wells were found blocked in the WH section or needed calibration for the adjustable choke. The GC-n pre-cleaning Production was: 130,445 BOPD and Post-cleaning Production was: 133,672 BOPD. The blockage rectification resulted in around 3,500 BOPD production increase.

From the field checks, three problems findings categories were observed. They are elaborated in the next subsections.

Finding Type 1 - Genuine Scale Blockage

Wells with typical finding of genuine scale blockage were highlighted as blocked in the WH BIP and confirmed by the field operation team. The common blockage material seen was scale and mixed with oil. Figure 4 shown the observation for a well example falling into this category. These wells choke valve and the spool section were cleaned accordingly and after which, the WHP returned to lower value, closer to the straight trend line in BIP.

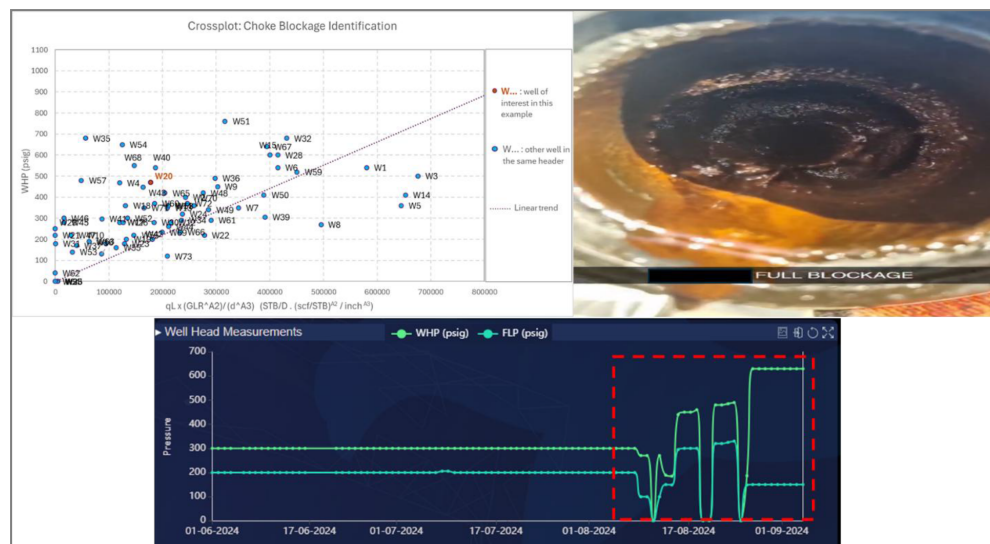


Figure 4—Example of Genuine Scale Blockage Finding Clockwise from Top Left: (a) WH BIP indicating the Suspected Well, (b) Photo From Physical Bean Check, (c) Historical Trend of The Well Indicating WHP increase - Likely the Blockage Forming.

To further verify the theory, another well was picked. This time the picked well was following the trendline of the BIP which should indicate clear unblocked wellhead and flowline. The well with data given in Figure 5 was checked by the field operation team and indeed was found clear, confirming the BIP finding.



Figure 5—Example of No Blockage Indication Verification Clockwise from Top Left: (1.) FL BIP Indicating Clean Condition, (2.) WH BIP Indicating Clean Condition, (3.) Historical Trend of The Well Indicating steady WHP and FLP.

Finding Type 2 - Adjustable Choke Miscalibration

Some wells appear to be highlighted in the BIP as blocked in the choke, but while dismantling the choke valve in the field, their internal parts were found clear. However, when carefully examining the wells' adjustable choke valve, they were observed to have calibration issues.

For example, in Figure 6, during the production test of the wells, the portable test separator choke was set at choke 42/64. Post the production test separator rigged down, the well's adjustable choke was set following the same 42/64 opening. Yet that adjustable choke dial was miscalibrated, hence was actually giving smaller opening and higher WHP. This was evidenced when the well's adjustable choke set at full closure position, that adjustable choke dial was reading 33/64. This means the well was actually flowing at choke 9/64 resulting in lower production rate than the production test.

The excessive choking acts as a blockage and is able to be detected by the WH BIP. All wells with this problem during the verification campaign were rectified by correctly calibrating the adjustable choke dial and setting the openings accordingly.

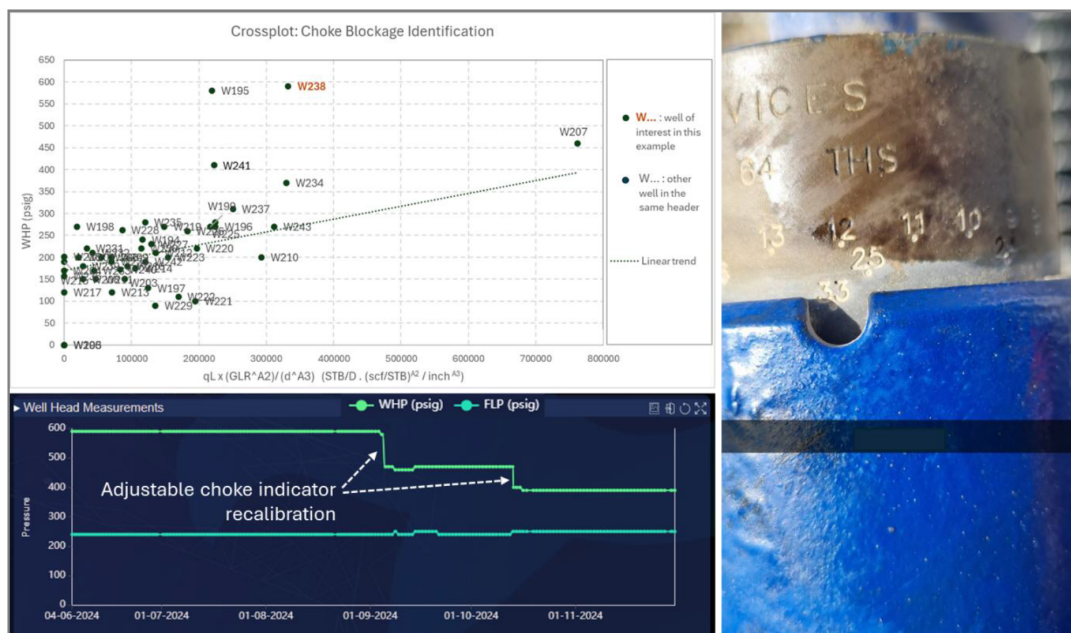


Figure 6—Example of Adjustable Choke Miscalibration Clockwise from Top Left: (a) WH BIP Indicating Blocked Condition, (b) Field Verification Found the Indicator Reading of 33/64 at the Adjustable Choke Full Closure, (c) WHP Trend Indicating Reduction of the Excessive Choking After Several Recalibrations.

Finding Type 3 - Emulsion Issue Requiring Chemical Injection Dosage Optimization

Figure 7 gives an example of a well highlighted as blocked in the FL BIP, yet when checked it was found clear from scale deposition. When reviewing the well and its fluid, it was found that the well's fluid appeared to be emulsion. The well history also shows some emulsion sample history where the well has been equipped with chemical injection pump injecting demulsifier chemical.

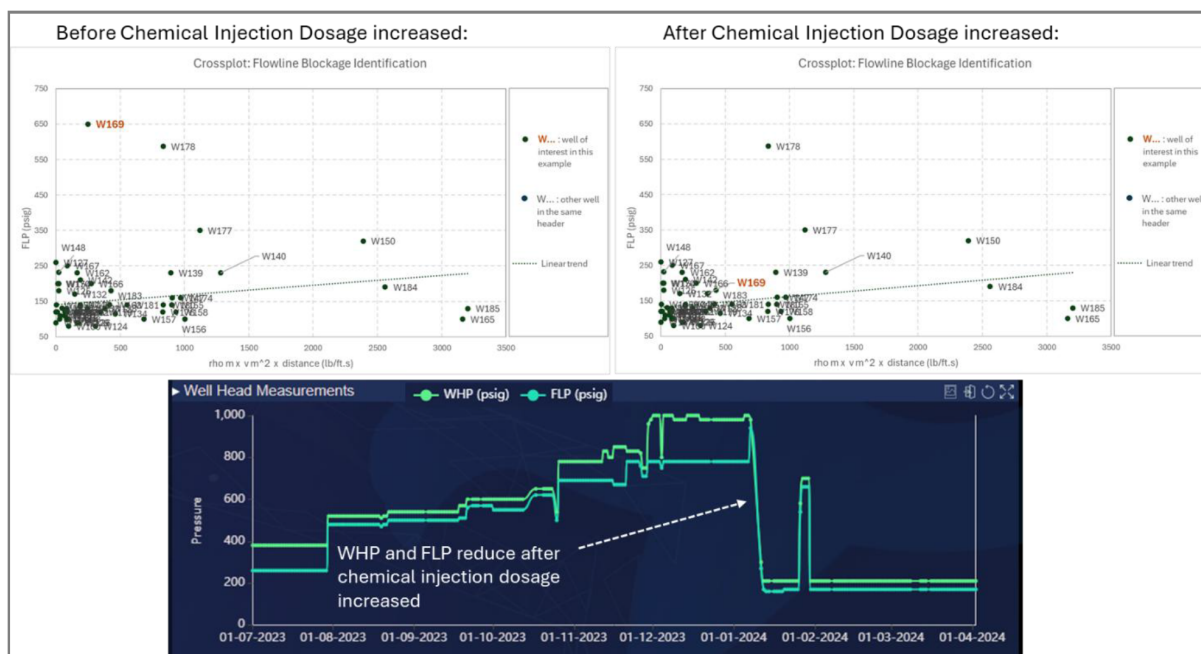


Figure 7—Example of Chemical Injection Dosage Optimization Clockwise from Top Left: (a) WH BIP Before Chemical Injection Increased Indicating Blocked Condition, (b) WH BIP After Chemical Injection Increased Indicating Normal Condition, (c) Well Pressure Trend Showing Response to the Chemical Injection Optimization.

With the well showing high pressure, it seemed like the demulsifier could be performing under optimum. We increased the demulsifier chemical dosage and found the well's FLP reducing to a significantly lower value. Replotting the value into the BIP indicated the data now lies in the normal trendline. This gives confidence that the chemical optimization works accordingly.

Summary and Conclusion

The BIP method has been validated in field and able to quickly screen wells with flow assurance related blockages, as well as those caused by choke miscalibration. BIP has brought great results in increasing the overall production for GC-n, whereby following-up through the necessary field rectifications onto the suspected BIP screened wells, the work contributed around 3,500 BOPD of restoration. The BIP technique helps to detect anomaly faster among the thousands of producer wells, helping to avoid potential production loss more efficiently. This is an improvement to the production loss detection with an aim to improve and even increase GC production. More will be expected in the future with continuous application of this approach using BIP. Without this practical tool, manual identification would result in delayed blockage identification due to the huge number of wells and limited number of staff as stated earlier.

This technique is also currently integrated into a "Blockage Elimination Task Force Team" workflow. This task force aims to tackle production losses due to flow assurance blockage occurrence in the flowlines and wellhead. With the BIP technique application, a fit-for-purpose and precise screening process of suspected blocked wells can easily be performed. This results in greatly increasing the rectification gain as the problematic well screening coverage is greater and able to be achieved in a shorter period as opposed to the full manual process.

Acknowledgements

The authors would like to thank KOC leadership and Government of Kuwait for granting permission to publish this material.

Nomenclature

- A1 = Constant used in two-phase empirical choke flow equation (Gilbert, Ros, Baxendell, or others)
- A2 = Constant used in two-phase empirical choke flow equation (Gilbert, Ros, Baxendell, or others)
- A3 = Constant used in two-phase empirical choke flow equation (Gilbert, Ros, Baxendell, or others)
- BIP = Blockage Identification Plot
- CIP = Chemical Injection Pump
- D = Pipe diameter (inch)
- d = Choke diameter (inch)
- D/S = Downstream
- f = Friction factor
- FL = Flowline
- FLP = Flowline pressure (psi)
- g = Gravity acceleration (32.2 ft/s²)
- GC = Gathering Center
- GLR = Gas Liquid Ratio (scf/STB)
- L = Pipe length (ft)
- P_{HEADER} = Header pressure (psi)
- QL = Liquid rate (STB/D)

subscript f = Friction component
subscript g = Gravity component
vm = Fluid mixture velocity (ft/s)
WH = Wellhead
WHP = Wellhead flowing pressure (psi)
 ΔL = Distance (ft)
 ΔP = Pressure drop (psi)
 ρ_m = Fluid mixture density (lb/ft³)

References

- Al Jadi, I., Desai, S., et al 2019. A Truly Integrated Wells, Flowline Networks and Process Facilities Model for the Greater Burgan Oilfield. SPE Kuwait Oil & Gas Show and Conference, Kuwait, October, 2019. SPE-198014-MS. <https://doi.org/10.2118/198014-MS>.
- Almohammad, H., Al-Saqabi, M. 2016. Production Surface Network Modelling and De-Bottlenecking Analysis for Gathering Center in Kuwait. SPE Asia Pacific Oil & Gas Conference and Exhibition, Perth, Australia, October 2016. SPE-182226-MS. <https://doi.org/10.2118/182226-MS>.
- Beggs, D. 1991. Production Optimization Using Nodal Analysis, second printing. Tulsa, Oklahoma: Oil & Gas Consultants International Inc. Tulsa
- Economides, M. Hill, D. Ehlig-Economides, C. 1993. Petroleum Production System, first edition. Englewood Cliffs, New Jersey: PTR Prentice Hall
- The Global Historical Weather and Climate Data. 2025. Ahmadi, Kuwait Climate, Ahmadi, KW Climate Zone, Monthly Weather Averages and Historical Data (accessed 27 May 2025)